

Lucía Cabrera

luciacabrera.ar



Pick a flower on Earth and
you move the farthest star.
Paul Dirac

Personal

Lucía Cabrera
nationality: Argentinian
born: 20/03/1991

Areas of specialization

- Theoretical Physics
- Mathematical Physics
- String Theory • M-Theory

Languages

Spanish	C2	mother tongue
English	C2	
German	B2	
Italian	B2	

Programming

L ^A T _E X	
haskell	
python	

Other Interests

- Functional Programming
- Quantitative Research
- Structured AI Reasoning
- Renaissance Music
- Philosophy



EDUCATION

- 2021 – **Physics Ph.D. Student**
BALSEIRO INSTITUTE · Bariloche, Argentina
Thesis: Particle Physics, Quantum Gravity and Dualities in String Theory.
Supervisor: Prof. Gerardo Aldazabal **Co-supervisor:** Prof. Bobby Acharya
- 2019 **Physics Licentiate Degree**
NATIONAL UNIVERSITY OF ROSARIO · Rosario, Argentina
Thesis: Dual Holographic Description of Yang-Mills Plasmas **Supervisor:** Prof. Martin Schvellinger (La Plata Institute of Physics - National University of La Plata). **GPA:** 9.53/10

MANUSCRIPTS

- In preparation **On Capturing Stringy Features within a Field Theory Description**
G. ALDAZABAL, E. ANDRÉS, L. CABRERA, M. MAYO ·
Embedding stringy features into a QFT and its hidden symmetries.
- In preparation **Branches of M-Theory/Heterotic string theory duals**
B. ACHARYA, D. BALDWIN, L. CABRERA, K. NARAIN ·
A classification of Heterotic string theories dual to M-theory compactified on a class of G_2 manifolds.
- 2026 **Too long; didn't solve**
L. CABRERA, I. SAXTON-KNIGHT ·
On how prompt length shapes LLM behaviour and problem discriminativeness.

HIGHLIGHTED SCHOLARSHIPS & AWARDS

- 2023 **STEP Fellowship**
INTERNATIONAL CENTRE FOR THEORETICAL PHYSICS · Trieste, Italy
Tri-annual international research collaboration fellowship.
- 2020 **Highest GPA in Physics**
PUNTO BIZ MAGAZINE · Santa Fe, Argentina
Highest GPA in a physics degree in the province of Santa Fe, Argentina.
- 2019 **Doctoral Fellowship**
NATIONAL SCIENTIFIC AND TECHNICAL RESEARCH COUNCIL · Argentina
- 2019 **Summer Research Scholarship**
BALSEIRO INSTITUTE · Bariloche, Argentina
Research internship. **Project:** Communication Servers for LAHN/ANDES.
- 2018 **Scientific Vocations Stimulus Scholarship**
NATIONAL INTER-UNIVERSITY COUNCIL · Argentina
Research fellowship. **Project:** Geometric Phases in Quantum Physics.

WORK EXPERIENCE

- 2025 – **AI Evaluation & Benchmarking – Maths & Physics**
Poindexter Labs.
- 2022 – 2025 **Teaching Assistant**
Balseiro Institute · Mathematics I & II.
- 2020 **Freelance Technical Writer**
- 2016 – 2019 **Teaching Assistant**
National University of Rosario · Mathematics I & II, Numerical Analysis.
- 2016 **Museum Guide**
Astronomy Complex of Rosario.

INVITED TALKS

- 2024 **Invited Talk** · Imperial College London. United Kingdom.
- 2024 **Contributed Talk** · Siembra - HoLAGrav 5th Young Frontiers Meeting. Online.
- 2024 **Contributed Talk** · 1st Online Conference for Women in Theoretical Physics from LATAM. Online.

RESEARCH VISITS

- 2025 **ICTP** - as STEP fellow. Italy.
- 2024 **ICTP** - as STEP fellow. Italy.
- 2023 **ICTP** - as STEP fellow. Italy.
- 2022 **IPhT Saclay**. France.